When Structured Conditioning Helps: What Each Mechanism Can Reach, and What None Can

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Abstract

Conditioning a network means injecting a class, an action or a caption into its computation, and three mechanisms carry almost all of it: concatenation, feature-wise modulation (FiLM), and hypernetworks. Which to use is settled by sweeping, and the resulting tables disagree across domains. We show the choice is partly decidable in advance, because each mechanism reaches a different set of feature maps and the polar decomposition says which. Approximation error then has a closed form per mechanism, and two consequences are impossibilities rather than tendencies. No orthogonal conditioner can represent a transformation with an eigenvalue of modulus $\neq 1$ in *any* representation, so conditioning that must specify content is structurally out of reach for it. And exact composition forces linearity while content forbids it, so the two roles cannot share a channel; the minimal algebra carrying both is $\mathbb{C}^* \cong \mathbb{R}_{>0} \times U(1)$, which derives magnitude-times-phase modulation rather than proposing it.

We test the account with two operators it names, a group-structured conditioner with exact composition at FiLM cost and its complex extension, under 25 pre-registered gates that returned 15 negatives. The predictions hold where the account says they should: the advantage is +0.6% on one condition and +46.6% on four (p=9.1×10^{-5}) and absent at weak single conditions, so it is composition rather than difficulty; the polar type of a Navier-Stokes operator named which of our own arms would fail before we understood why; and content conditioning fails by 8×, as the impossibility requires.

The account's empirical half is weaker, and we report its failure. Two short runs are meant to predict which tasks pay. Both passed on a sixteen-control chain we built because the account favored it, and the advantage was flat at −1.1%. A check already in our repository had called that outcome. Structured conditioning pays when a task's own conditions compose the way the operator does, which no dimensional criterion captures and which our screen did not measure.

1 Introduction

Nearly every conditional model must inject something into its computation: a class, an action, a timestep, a text embedding. Call this conditioning. Diffusion models modulate every block on the timestep and class via adaLN [2]; visual reasoning, speech and RL systems modulate features on questions, speakers and goals [1]; world models condition latent dynamics on actions. Three mechanisms carry almost all of it: concatenation, feature-wise affine modulation (FiLM), and a hypernetwork that emits weights from the condition.

Which one to use is settled empirically. Papers train all three and print a table, and the tables disagree across domains. No account says which mechanism suits which task, so every new setting repeats the sweep, and a practitioner who reads two papers gets two answers. Geometric deep learning explains when symmetries of the *data* help, but conditioning is a different object: whatever structure exists lives in the condition space.

The choice is less open than the sweeping suggests, because the mechanisms do not differ in how much they can express so much as in *what* they can express. A condition names one of two things, and the distinction runs through this paper: it can name a *change* to apply, such as a pose offset or an edit, or the *content* to produce, such as a class or a caption. Those are different operations on the features. Every invertible map factors uniquely into a rotation and a stretch, and each mechanism reaches essentially one factor: FiLM the stretch, a rotation the rotation, a hypernetwork both. Approximation error is then not a matter of headroom but of class membership, and it has a closed form (§3).

Two consequences are impossibilities. An orthogonal conditioner cannot represent a transformation with an eigenvalue of modulus $\neq 1$ in *any* representation, because a change of basis is a conjugation and conjugation preserves the spectrum; content conditioning needs exactly such eigenvalues, so it is unreachable rather than difficult, and no amount of co-training repairs it. Separately, exact composition forces the condition-to-modulation map to be linear, by

Cauchy's functional equation, while content is not additive and forbids linearity. The two roles therefore cannot share a channel, and the smallest algebra carrying both is $\mathbb{C}^* \cong \mathbb{R}_{>0} \times U(1)$: magnitude and phase. That derives a conditioner rather than proposing one.

We test the account with the two operators it names (§4). Both already exist in special cases: RoPE [4] and GRAPE [5] encode position as a group action $G(n) = \exp(n\,\omega L)$, exactly compositional because position enters linearly in the Lie algebra. Carrying that construction from scalar positions to arbitrary condition vectors in FiLM's role gives **CGA**, and extending its algebra gives **Complex FiLM**. Twenty-five pre-registered gates returned fifteen negatives, and the wins and the negatives land where the account says. Composition is confirmed by shape rather than by margin: the advantage is 0.6% on one factor and rises to 25.5%, 38.7% and 46.6% on two, three and four, while every arm stays within 1.6% of the others at weak single conditions, so it cannot be difficulty or strength (Figure 1). Content conditioning fails by 8×, as the impossibility requires. The polar type of a Navier-Stokes operator named which of our own arms would fail before we understood why.

The account's empirical half is weaker, and reporting its failure is part of the contribution. Two short runs are meant to decide whether structure pays. We built the case the account favored, a sixteen-control parametric chain, and predicted a win before running it; the advantage was flat at -1.1%. A check already in the repository had called that outcome, and it measures the condition the screen was missing: whether the task's own conditions compose the way the operator does.

Contributions. (1) What each mechanism can reach. The reachable sets of FiLM, rotations, hypernetworks and complex modulation are nearly disjoint rather than nested, so parameter count is the wrong currency; the polar decomposition gives each one's approximation error in closed form (§3). **(2) Two impossibilities.** No orthogonal conditioner represents an eigenvalue of modulus $\neq 1$ in any representation, and no conditioner carries content and exact composition on one channel. The second derives magnitude-times-phase modulation as the minimal resolution, and shows it is maximal (§3). **(3) Two operators that make the account testable.** A group-structured conditioner with composition exact for any weights, recovering FiLM, RoPE and GRAPE as special cases, and its complex extension at 0.84× FiLM's cost (§4). **(4) Evidence, including fifteen negatives.** A pre-registered, budget-fair protocol over 25 gates, with margins, tests and single-read splits committed before every run. Composition is established by shape rather than margin, and we report two verdicts that cleared their criteria as failures because the criteria named the wrong baseline (§5, §6). **(5) The account's own falsification.** Its two screening quantities predicted a win on the one task where the prediction preceded the run, and the advantage was flat at -1.1%. We report the missing condition and the instrument that measures it (§7).

2 Related work

FiLM and conditional normalization. FiLM [1] and its normalized variants (conditional BN/LN, adaLN in DiT [2]) apply per-feature affine transforms generated from a condition. All are diagonal. We keep FiLM's role and enrich only the operator.

Structured operators in parameter-efficient fine-tuning (PEFT). Low-rank, orthogonal, and butterfly operator families are well studied as weight fine-tuning: LoRA [6], OFT [7], BOFT [8], HRA [9], Group-and-Shuffle [10]; He et al. [11] give a unified view. These learn one fixed transform per task. Our setting differs on every axis that matters: per-sample, input-conditioned, activation-space, and evaluated for compositional generalization. Hypernetworks [3] generate weights from conditions and inherit the memorization behavior we quantify.

Position encodings as group actions. RoPE [4] rotates query-key pairs by angles linear in position; GRAPE [5] generalizes this to group representations with exact relative laws. The mechanism is theirs; our contribution is the transplant to arbitrary condition vectors in FiLM's role, with a learned basis, and the budget-fair compositional evidence.

Compositional generation. Montero et al. [12] showed reconstruction-trained generative models fail on unseen factor combinations. Score and energy composition [23] composes concepts across separate conditional passes; we compose conditions inside the operator, in one pass. Symmetry-based representation learning [13, 14] argues latents should carry group actions; we supply the conditioning-side mechanism.

Nearest prior work. We searched deliberately for work that would invalidate this paper. The closest hits: Feature-space rotations with exact composition exist for geometric transformations [17]; abelian exponential-map operators for disentanglement [22]; attribute operators for recognition [26]; multilinear attribute mixing for unseen combinations [27]; per-sample amplitude-and-phase activation modulation in periodic implicit networks [20]; and composition by complex phase addition in knowledge-graph embeddings [21]. Rotation-structured latent world models with latent-prediction losses originate with Quessard et al. [15], Caselles-Dupré et al. [16], and the homomorphism autoencoder [18], whose composition property is emergent rather than imposed; latent consistency objectives are standard in model-based RL [19]. Distillation moves guidance into the conditioning path [25], and prior work documents CFG's high-scale distortion [24].

We claim none of these components, and several of them are instances of what we analyse: the amplitude-and-phase modulation of Chan et al. [20] and the phase composition of Sun et al. [21] are the split our §3 derives, arrived at by construction rather than by necessity. What is missing from all of it is an account of *which* mechanism suits which task. The operator families above are studied as parametrizations and compared by benchmark; none states what a mechanism can and cannot represent, and none predicts its own failures. That account, its two impossibilities, and the pre-registered programme that tests and partly falsifies it are what we contribute.

3 Which maps a mechanism can reach

Mechanism choice looks like a matter of taste because the usual currency, parameter count, does not measure the thing that decides it. This section replaces that currency.

Write a conditioning mechanism as a family of feature maps $\{T_c\}$ indexed by the condition, where FiLM contributes the diagonal maps, a hypernetwork all linear maps, and the two operators we build in §4 the orthogonal maps (**CGA**) and the block-diagonal scaled rotations (**Complex FiLM**). Split test error into an approximation term, an estimation term and an extrapolation term, $E_{\text{approx}} + E_{\text{est}} + E_{\text{extrap}}$. Two facts follow, and one of them forecloses a class of designs outright. Full derivations are in the released `THEORY.md`; §6 measures every quantity previewed here.

Approximation error is class membership, not capacity. The reachable sets are nearly disjoint rather than nested: the diagonal maps and the orthogonal maps intersect only in the 2^d matrices with entries ± 1. Switching from FiLM to a rotation does not shrink the reachable set, it *moves* it, so parameter counts are the wrong currency. Complex FiLM uses 0.85× FiLM's parameters while substituting one operation class for another. The right coordinates are polar, $T = US$, and each mechanism reaches one factor. This yields exact minimisers, not bounds: $E_{\text{approx}} = 0$ for a hypernetwork, $\|T^* - \operatorname{diag} T^*\|_F$ for FiLM, $\|S^* - I\|_F$ for a rotation (orthogonal Procrustes), and a blockwise projection residual for Complex FiLM. So a rotation's approximation error *is* the stretch content of the task's operator, which is why an isometry fails at content by 8× at any parameter count (§6).

One impossibility, and its converse. If the required transformation has any eigenvalue of modulus $\neq 1$, no orthogonal conditioner can represent it in *any* representation: a change of basis is a conjugation, and conjugation preserves the spectrum. Content conditioning needs $|\lambda| \to 0$, so its 8× failure is structural rather than incidental. The converse matters as much. Approximation error refers to the transformation required *in the coordinates the encoder produced*, so it is a property of the task and representation together, not of the task alone. A learned representation can repair a rotation wearing the wrong basis and can never repair a change of scale. This predicts that freezing the representation is fatal, which §6 confirms, and it carries a design rule: the operator must be the *only* conditioning path, since a parallel unstructured route removes the pressure to find such coordinates. Our own hybrid arm, which contained a concatenation path, scored +1.7% against concatenation alone, a tie.

Exact composition bounds extrapolation, and only on the compact factor. For $T(c) = \exp(A(Wc))$ with A skew, the exponential is 1-Lipschitz, giving $\|\hat{T}(c) - T(c)\|_F \leq \|A\|\,\|\hat{W} - W\|\,\|c\|$. The bound depends on $\|c\|$ alone and not on how far c sits from the training conditions: reaching an unvisited *combination* costs nothing beyond reaching the same magnitude, and if the training conditions span the condition space then W is identifiable and the error vanishes with data. With a symmetric part present the Lipschitz property fails and error grows like e^s, which is why the magnitude channel needs a clamp while the phase channel composes to 10^{-6}, and why guidance degrades linearly in strength on the rotation channel and exponentially on the magnitude one.

Complex FiLM is forced, not chosen. The two conditioning roles cannot share a channel, and the algebra says what to do instead. Exact composition requires $A(c_1{+}c_2) = A(c_1) + A(c_2)$ in algebra coordinates, and a continuous additive map is linear by Cauchy's functional equation: there is no nonlinear exactly-compositional conditioner. Content runs the other way. Specifying what to produce is not additive, since composing "cat" with "dog" is not "cat + dog", so the condition-to-modulation map has to be expressive; the fully linear variant scores 0.19 in the content role against the expressive variant's 0.97 (§6). The two requirements are therefore incompatible on one channel, which makes splitting the representation a theorem rather than a design preference. The smallest algebra carrying both an expressive and an exactly-compositional direction is $\mathbb{C} = \mathbb{R} \oplus i\mathbb{R}$, whose multiplicative group factors as $\mathbb{C}^* \cong \mathbb{R}_{>0} \times U(1)$: magnitude and phase. Compactness fixes which carries which. Rotations are bounded, so exactness survives on them; e^s is unbounded, needs a clamp, and a clamp is nonlinear. We measure composition error of 10^{-6} while $|s| \leq 4$ and order 1 the moment the clamp engages. The construction is also maximal: two dimensions per 2×2 block over $d/2$ blocks is d, the dimension of a Cartan subalgebra of $\mathfrak{gl}(d)$, so no exactly-compositional linear conditioner reaches a larger family. Going further costs commutativity, and with it exactness at arbitrary weights.

4 Two conditioners the account names

The account of §3 names two operators: one that reaches the rotation factor exactly, and one that reaches both factors at the cost of exactness on half of them. Both already exist in special cases, and writing them in this form is what makes them testable.

Let $x \in \mathbb{R}^d$ be features and $c \in \mathbb{R}^k$ a condition. FiLM computes $y = \gamma(c) \odot x + \beta(c)$; hypernetworks emit $W(c)$. All are instances of

$$y = T(c)\,x + \beta(c), \tag{1}$$

and they differ only in where the per-sample operator $T(c)$ comes from.

Definition 1 (Compositional conditioner)**.** *$T : \mathbb{R}^k \to \mathrm{GL}(d)$ is* compositional *if it is a homomorphism from $(\mathbb{R}^k, +)$: $T(c_1{+}c_2) = T(c_1)\,T(c_2)$ and $T(0) = I$.*

Proposition 1 (Exact composition by construction)**.** *Let $\mathfrak{a}$ be an abelian subalgebra of $\mathfrak{gl}(d)$, $\mathrm{A} : \mathbb{R}^m \to \mathfrak{a}$ linear, $W \in \mathbb{R}^{m\times k}$, and P orthogonal. Then $T(c) = P \exp(\mathrm{A}(Wc))\, P^\top$ is a compositional conditioner for every W, P. If $\mathfrak{a}$ is skew-symmetric, $T(c)$ is orthogonal.*

Proof. $\exp(X)\exp(Y) = \exp(X{+}Y)$ for commuting X, Y; $\mathrm{A} \circ W$ is linear; conjugation preserves products; $\exp(0) = I$. □

Composition is usually a behavior the network must learn from data. Here it is a property of the parametrization. It holds at initialization, during training, and far outside the training distribution. Training only decides which directions the condition rotates and how fast.

The guarantee is conditional on coverage, and the condition is easy to miss. Composition determines T at an unseen condition only when the training conditions generate that condition under the group operation. Training on conditions that do not generate the tested ones buys no extrapolation, however exact the homomorphism, which is why a large compositional gap can indicate a test set outside the training set's reach rather than an opportunity.

Special cases. Choices of $(\mathfrak{a}, c, P)$ recover deployed mechanisms:

$\mathfrak{a}$	condition	P	recovered mechanism
$\mathrm{diag}(\mathbb{R}^d)$	c	I	*exponential FiLM*: $y = e^{Wc} \odot x$ (compositional)
$\mathfrak{so}(2)^{d/2}$	$c = n$ (scalar position)	I	RoPE [4]
$\mathfrak{so}(2)^{d/2}$	$c = n$, learned planes	learned	multiplicative GRAPE [5]
$\mathfrak{so}(2)^{d/2}$	general $c \in \mathbb{R}^k$	structured	**this work** (FiLM's role)
$\mathrm{diag} \times \mathfrak{so}(2)^{d/2}$	general c	canonical	**Complex FiLM** (§6)

Standard FiLM reaches the same diagonal operators through an MLP head $\gamma(c)$. The operators match; only the composition law is missing, and §6 shows that its absence is what breaks FiLM on unseen combinations.

Cost. With $\mathfrak{a} = \mathfrak{so}(2)^{d/2}$, exp is closed-form planar rotation: $3d$ FLOPs, no matrix exponential. The basis P is the whole budget question. A dense learned P costs $4d^2$ FLOPs per sample, which is 1.52× FiLM's entire conditioning path at d=128. Two fixes work. A structured orthogonal P (five block-diagonal Cayley layers with fixed shuffles, following Group-and-Shuffle) brings the total to 1.11× FiLM, with 4,800 parameters against the 8,128 dimensions of SO(128); that is enough, because P only needs the planes the condition acts on. Inside a transformer no explicit P is needed, since the adjacent dense projections absorb it exactly as RoPE relies on the query and key projections rather than on a basis of its own. Linearity also gives a strength dial for free: $T(c)^w = T(wc)$, so raising the operator to a power is the same as scaling the condition, which §6 uses in place of CFG's extrapolation between a conditional and an unconditional prediction. Unit tests pin identity at initialization and at c=0, orthogonality, and exact composition. Complex FiLM extends the algebra to diag × $\mathfrak{so}(2)^{d/2}$: each feature pair is multiplied by $m\, e^{i\theta}$, with magnitude from FiLM's expressive head (the content channel) and phase linear in the condition (the composition channel), at 0.84× FiLM's cost in our harnesses.

5 Pre-registered, budget-fair protocol

Vocabulary. An *arm* is one conditioning method, trained under the shared recipe. A *suite* is one experiment: a fixed backbone, dataset, and training loop that all of its arms share. A *gate* is that suite's pass-or-fail rule, written down before the run as a conjunction of *criteria*: typically a margin over the strongest baseline on unseen combinations, a no-regression criterion on the training distribution, and a budget criterion. A suite passes only if all of its criteria pass, and we report the outcome either way.

Design. We commit every suite's success margin, statistical test (N=10 seeds, one-sided Mann–Whitney $p \leq 0.01$, Cliff's $|\delta| \geq 0.474$) and splits to version control before its decision run. Each test split is read once, after all selection on a validation split. All arms share one backbone and one training recipe. A shared counter enforces the budget: our conditioning path stays within 1.20× FiLM's FLOPs and 1.05× the smallest hypernetwork baseline's parameters, while baselines reach 48× our parameter count. We record deviations as dated amendments in the committed specifications, plus one accounting erratum that our own audit caught and that downgraded a favorable verdict.

Threats to validity, and the design answer. The benchmarks are self-constructed, so we built them to remove judgment from scoring: dSprites and 3D Shapes are deterministic factor-to-image maps, so every held-out condition has a unique ground-truth image and pixel error is an objective compositional metric, with no perceptual score and no human rating. Pre-registration removes outcome-contingent analysis, and fifteen of the twenty-five verdicts below are negatives, one of which falsified our own headline prediction. Every number here regenerates from the committed logs with one command per suite.

Arms. FiLM; conditional LayerNorm; Concat-MLP; a dense hypernetwork $I{+}\Delta W(c)$; a dynamic low-rank map $I{+}A(c)B(c)^\top$; the Lie operator; Complex FiLM (both variants); and an MLP-head ablation of our own method (same basis, same operator family, more compute, no linearity).

The experiments. **Two synthetic suites.** Learn $y = M(c)x$, where c picks a combination of 8 rotation primitives. The second hides the primitives behind a random basis, so the operator has to find them, and additionally tests all 56 never-trained triples. **Three image suites** on dSprites and 3D Shapes. Encode a real image, apply $T(\Delta)$ to the learned latent for a factor change Δ, decode, and score pixel error against the true transformed image; OOD means never-trained *types* of two-factor change. Two of the three add a categorical shape factor. **Content.** A small diffusion transformer generates dSprites images from a full factor specification, so the condition names what to draw rather than how to change it. **Rollouts.** A world model applies the conditioning arm recurrently, one action per step, tested at horizons 10 and 20 after training on 3. **The successor.** Complex FiLM, through the dSprites and content setups. **Guidance.** Condition-dropout training, then classifier-free guidance at strength w for FiLM against condition powering for Complex FiLM, scored against deterministic ground truth at each strength.

Table 1: Error on never-trained condition combinations: mean (seed sd) over 10 seeds. CGA is lowest in every column, with complete seed separation against the strongest hypernetwork baseline (p=9.1×10^{-5}, δ=−1.0). OOD means out-of-distribution: a condition combination never seen in training. Every column except 3D Shapes passed its full pre-registered gate; 3D Shapes failed on a separate training-fit criterion (text). The last column reruns the dSprites setup with Complex FiLM. **Columns are not comparable to one another**: the two synthetic suites report operator MSE, the rest report per-pixel MSE.

Conditioner	Aligned	Hidden basis	dSprites	+shape	3D Shapes	Complex FiLM
FiLM	1.3 (0.1)×10^{-2}	1.6 (0.04)×10^{-2}	3.0 (0.2)×10^{-3}	5.9 (0.2)×10^{-3}	4.5 (1.0)×10^{-3}	3.1 (0.2)×10^{-3}
Concat-MLP	1.5 (0.3)×10^{-2}	1.1 (0.4)×10^{-2}	2.9 (0.1)×10^{-3}	6.3 (0.2)×10^{-3}	5.4 (5.2)×10^{-3}	–
Cond-LN	2.4 (0.1)×10^{-2}	2.7 (0.05)×10^{-2}	3.2 (0.2)×10^{-3}	6.1 (0.3)×10^{-3}	8.6 (13.1)×10^{-3}	–
Hypernetwork	1.9 (0.3)×10^{-3}	3.4 (0.9)×10^{-3}	3.8 (0.3)×10^{-3}	7.0 (0.4)×10^{-3}	9.3 (2.3)×10^{-3}	3.8 (0.2)×10^{-3}
Dyn. low-rank	5.7 (0.5)×10^{-3}	6.1 (0.4)×10^{-3}	5.1 (0.9)×10^{-3}	8.1 (0.4)×10^{-3}	2.0 (0.4)×10^{-2}	–
MLP-head (abl.)	–	1.1 (0.2)×10^{-3}	2.7 (0.2)×10^{-3}	5.0 (0.2)×10^{-3}	3.3 (0.8)×10^{-3}	–
Complex FiLM (linear mag.)	–	–	–	–	–	2.0 (0.1)×10^{-3}
Complex FiLM	–	–	–	–	–	2.0 (0.2)×10^{-3}
CGA (ours)	7.4 (2.0)×10^{-4}	3.3 (2.2)×10^{-8}	1.8 (0.1)×10^{-3}	4.0 (0.1)×10^{-3}	1.2 (0.2)×10^{-3}	1.8 (0.1)×10^{-3}

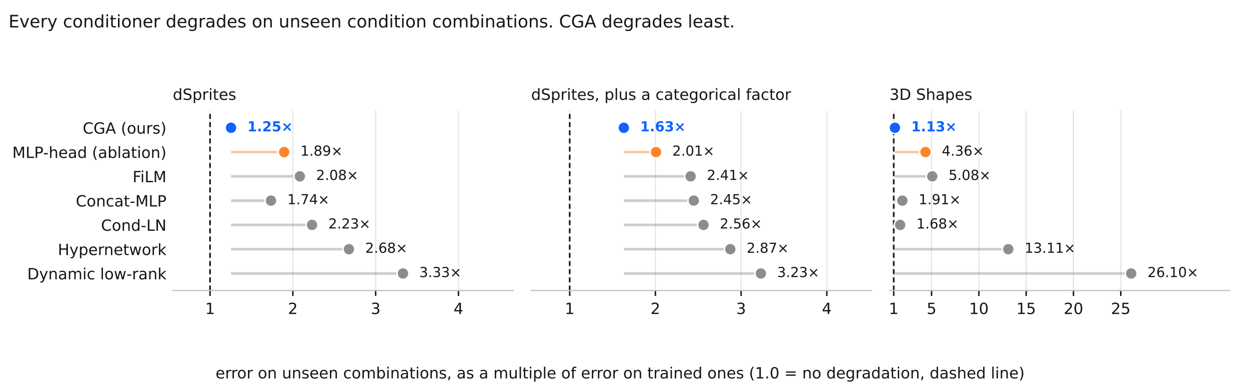

Figure 1: How much worse each conditioner gets when conditions are combined in ways it never saw, as a multiple of its own error on trained combinations. A value of 1.0, the dashed line, would mean no degradation at all. CGA sits closest to it on all three image suites, and the two conditioners that can express any linear map sit furthest.

6 Evidence: where the account holds

Each result below tests something §3 asserts. Results (1) to (3) test the composition claim and its mechanism, (4) tests the derived conditioner, (5) and (6) test the impossibility and the fit penalty it implies, and (7) to (9) test the representation, rollout and guidance consequences. Every verdict comes from a gate committed before its run.

(1) Composition is exact, and the weights say what each condition does. On the hidden-basis suite, CGA reaches 3.3×10^{-8} error on unseen pairs and 5.3×10^{-8} on never-trained triples. The best baseline sits five orders of magnitude higher, and every baseline degrades as combinations lengthen (Figure 3). Learned angles match the true generators to 2×10^{-4} rad, so reading the weights tells you which plane each condition rotates and how fast (Figure 4).

(2) The advantage comes from composing conditions, and its shape shows it. Comparing arms on a fixed evaluation cannot separate a compositional mechanism from an arm that is simply better. We therefore swept the number of simultaneously changed factors and their magnitude independently, and registered in advance that the advantage must vanish for a single weak condition and grow with the number composed. At the trained magnitude the operator gains 0.6% over the best additive arm on one factor, then 25.5%, 38.7% and 46.6% on two, three and four (p=9.1×10^{-5}, δ=−1.00); at weak single conditions every arm falls within 1.6% of the others. An advantage that is absent at one

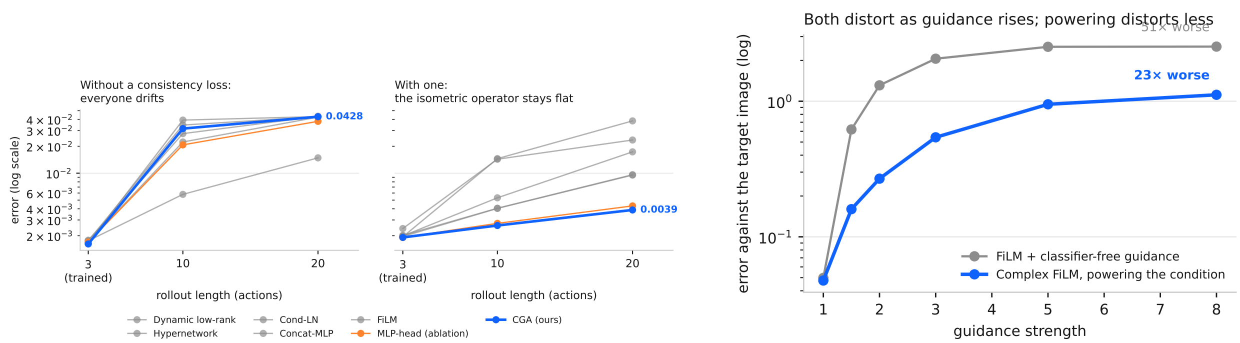

Figure 2: **Left:** error against rollout length, trained only to length 3. Without a consistency loss every conditioner drifts at much the same rate. Adding one separates them: the isometric operator flattens out while the baselines keep climbing. **Right:** what turning guidance up does to the generated image. Both distort, since the target here is deterministic and there is no distribution to sharpen, but classifier-free guidance distorts twice as fast.

condition and rises with the number composed cannot be difficulty, strength, or a generally stronger arm.

The magnitudes follow. On dSprites every arm fits the training distribution equally well, which isolates composition: CGA cuts error on unseen combinations by 53.8% against the strongest baseline at 39× fewer conditioning FLOPs, and by 42.9% once a categorical shape factor joins the condition. Moving from trained to unseen combinations costs CGA 1.25×, or 1.63× with the shape factor, against 2.1–3.3× for every baseline. We call that multiplier the *compositional gap*.

(3) The linear map is the mechanism, and the operation class rather than the parameter count explains the rest. The ablation keeps the basis and the operator family, spends more compute, and swaps only the linear map for an MLP. It loses by $4.3{\times}10^4\times$ on synthetic triples and 1.55× on dSprites. The two hypernetwork arms can express any operator and post the largest compositional gaps of any arm on every image suite, 2.7–3.3× on dSprites and 13–26× on 3D Shapes against CGA's 1.13–1.63×.

We first read this as capacity memorizing the training combinations, and our own audit does not support that reading. Sorting the arms by parameter count does not sort them by error: the 2.1M hypernetwork beats the 298K low-rank arm on dSprites (0.00383 against 0.00508) and on 3D Shapes (0.00928 against 0.01978), and the 83K Concat-MLP often beats the 50K FiLM. What survives is weaker and points elsewhere: the unstructured arms are usually worst on held-out combinations without being ordered among themselves, which implicates which maps an arm can express rather than how many parameters express them (§3).

(4) Complex FiLM covers both conditioning roles at lower cost. Complex FiLM multiplies feature pairs by $m\,e^{i\theta}$, giving content to the magnitude channel and composition to the phase. Under two pre-registered gates it beats FiLM by 36% on unseen change types at fit parity and 0.84× FiLM's cost, and matches FiLM on content generation at 0.97× its error (pre-registered non-inferiority margin 1.10×), where the pure rotation failed by 7×. Giving the magnitude channel a linear map instead of an MLP fails that same content metric (0.19 against FiLM's 0.04), so content needs the expressive head. FiLM is the magnitude channel on its own; the full complex multiplication does both jobs.

A third gate confirms it outside the disentanglement benchmarks. On a 2D Navier-Stokes surrogate conditioned on four physical parameters (viscosity, drag, forcing and background advection, with the initial vorticity field supplying all of the content), Complex FiLM passes every criterion on fresh seeds: 45.4% below FiLM, 64.6% below the best unstructured arm at $\delta{=}{-}1.00$, and an in-distribution fit ratio of **0.938**×, the first arm here to fit better than the unstructured baseline, replicated across independent seed sets. The claim covers Complex FiLM and not the pure group operator: its magnitude head is nonlinear in c, so composition stays exact on the phase channel only. It buys fit by spending half the guarantee, and the arm that keeps the guarantee on both channels failed this criterion twice.

(5) Content is provably out of reach for an orthogonal conditioner. In the content suite, rotation conditioning fails: pixel MSE 0.31 against FiLM's 0.04, an 8× regression, and the ablation fails identically. A proof explains it. Producing content means suppressing features, so the required transformation carries eigenvalues of modulus far from 1; every orthogonal matrix has all $|\lambda|=1$; and a change of representation acts by conjugation, which preserves the spectrum. No orthogonal conditioner can carry "generate this image" in *any* representation, and co-training cannot repair it (§3).

(6) The remaining fit penalty tracks content demand. Where the condition supplies some content, the operator fits worse in distribution than an unstructured arm. On 3D Shapes CGA posts the lowest error on unseen combinations (87.5% below the hypernetwork) and still trips the pre-registered no-regression criterion at 1.46×, and on the physics task the same criterion fails at 1.118×. Two registered relaxations that keep exact composition, a scaling generator and a learned diagonal conjugating the orthogonal basis, do not fix it: the conjugated operator looked like a cure on validation (0.962×) and reached 1.1007× on the held-out split, missing by 0.065 percentage points.

A magnitude channel does, but only where the condition supplies no content. Complex FiLM reaches 0.938× on the physics task, whose four parameters act on content the input already carries; on 3D Shapes, whose condition includes a categorical shape swap, it reaches 1.542×, worse than the pure rotation, while still passing the compositional criteria by 84.5%. The penalty tracks how much content the conditioning must supply. This family suits conditions that *transform* content supplied elsewhere, and no variant we tested rescues it otherwise.

(7) The advantage requires a co-trained representation. We ran the same dataset, held-out attribute pairs, arms and budget counter twice, varying whether the representation trains alongside the operator. We previously called this a single-variable comparison and it is not: the co-trained run optimizes a pixel objective and scores in pixel space, while the frozen run optimizes and scores in latent space, so representation, objective and metric all differ. The within-run ratios are the defensible part. Co-trained, CGA's error on unseen combinations sits 87.5% below the hypernetwork's. Frozen, against a plain autoencoder trained once for reconstruction and then shared unchanged by every arm, CGA lands 60.1% *worse* than that hypernetwork while using 48× fewer conditioning parameters, and fits 2.43× worse in distribution against a 1.10× ceiling. Complex FiLM fails there too, so the rotation channel is not the culprit.

The advantage is therefore a property of the operator and its representation together, and bolting it onto a frozen general-purpose encoder, the pattern latent-space editing deploys, loses. CGA still posts the smallest compositional gap of any arm that fits, 1.39× against the hypernetwork's 2.11×, so it degrades most gracefully and still loses: grace does not recover a fit handicap.

(8) Rollouts need a consistency loss, and the operator alone does nothing. In the rollout suite every arm compounds error at the same rate, because each step adds a small mismatch between the predicted and the true latent. A contracting operator shrinks past mismatches; an isometry preserves them, so they accumulate. Training with $\lambda\|T(a)z_s - z_{s+1}\|^2$, applied to every arm equally, inverts the picture: the rotation operator's horizon-20 error goes flat at 0.0039 (growth 2.0×) against the hypernetwork's 0.038 (growth 19.8×) and FiLM's 0.017 (Figure 2, left). The loss is standard in model-based RL [19], and rotation transitions with latent-prediction objectives appear in the homomorphism autoencoder [18]; giving every arm the same loss at the same budget shows that neither ingredient fixes rollouts alone. A null result sharpens the mechanism: shrinking the latent by a fixed factor each step, swept from 0.003 to 0.1, leaves horizon-20 error unchanged, so whatever removes the accumulated mismatch must depend on the state. A separate single-step criterion fails, so this suite counts as a negative in Table 1 despite the horizon-20 result.

(9) Guidance costs one forward pass, and holds only on a deterministic target. CFG's distortion at high scales is documented [24]; our benchmark isolates it against an exact alternative, because the conditioning target is deterministic. CFG's error grows 50.7× from strength 1 to 8, while powering the condition through the phase channel grows 23.7× (p=1.6×10^{-4}), stays 2–5× lower at every strength above 1, matches at native strength (0.96×), and needs no unconditional pass and no distillation [25] (Figure 2, right). Neither method improves over strength 1 here: the target is deterministic, so no distribution exists to sharpen, exactly as the suite's registered caveat stated.

The sweep in (2) refutes the other half of what we registered. We had argued that additive aggregation, which is what a value-sum over condition tokens performs, should degrade as $\frac{1}{2}\|\sum_i A_i\|^2$ and therefore worsen with conditioning *strength*. The advantage instead reverses beyond the trained scale, reaching −14.8%. Rotations are periodic, so past the trained angle the operator wraps and the error oscillates. The bound of §3 survives as an upper bound linear in $\|c\|$, but it is loose exactly where we relied on it. Exact composition buys accuracy when stacking conditions at the scale trained on, and gives no strength dial.

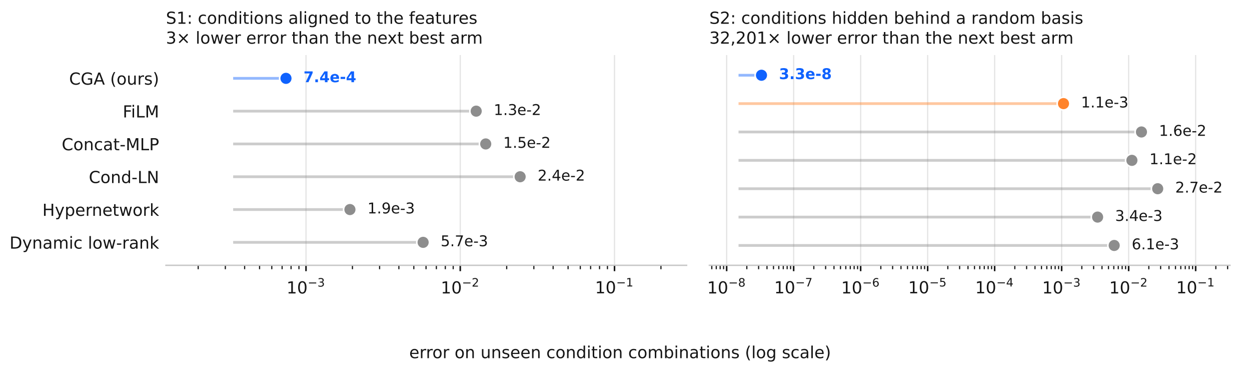

Figure 3: On the synthetic suites, where the conditions form a group exactly, the gap is not a matter of percentages. Each dot is one conditioner's error on unseen combinations; note the log axis. Hiding the conditions behind a random basis (right) costs CGA nothing, because it learns the basis, while the gap to the next best arm grows to five orders of magnitude.

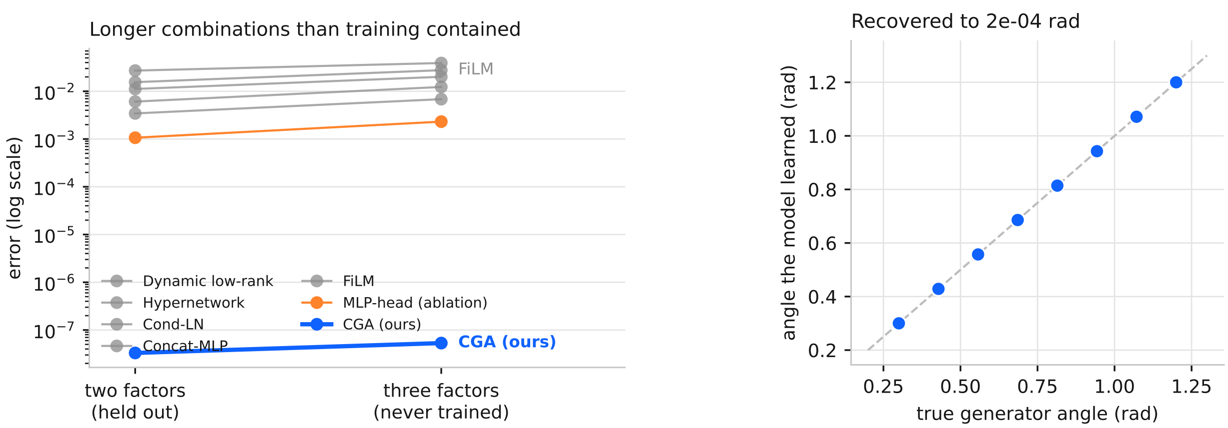

Figure 4: **Left:** what happens when a test combination is longer than anything in training. Every baseline gets worse; CGA does not move, because composition is a property of its parametrization rather than something it had to learn. **Right:** the angles the model learned for each condition, against the true generators of the data. Reading the weights tells you which plane a condition rotates, and how fast.

7 Turning the account into a screen, and watching it break

Two short runs estimate both terms. Choose the mechanism whose reachable factor matches the polar type the task requires; then adopt exact composition only if the evaluation visits unvisited conditions. Short runs estimate both quantities: in-distribution fit ratio for the first, compositional gap for the second. Table 2 applies both to every task we ran. They separate the wins from the losses, and each is independently observed to fail while the other passes: the audio chain had a fit ratio of 1.000 and no gap, PDEBench an enormous gap and an unusable fit. A one-factor account cannot produce that pattern.

The threshold is softer than the separation. Both quantities need a yardstick arm, and the two defensible choices disagree: measured against the baseline each gate nominated in advance, the Navier-Stokes Complex FiLM run fits at 0.938×, and measured against whichever baseline fits best, at 1.122×. The wins and losses stay separated under either, but the dividing line moves from 1.05 to roughly 1.20. We report the stricter yardstick throughout Table 2 and treat the threshold as fitted rather than derived.

The account also covers results the capacity framing could not. The synthetic Navier-Stokes operator is diagonal in Fourier, so its polar factors are computable rather than estimated: viscosity and drag are pure stretch (phase 0.000),

Table 2: The two screening quantities against the outcome, for every task we ran. The fit ratio is the tested arm's in-distribution error over the best-fitting baseline's; the gap is that baseline's held-out error over its own in-distribution error. Both are read off the baselines, since an arm cannot be its own yardstick. Predictions use fit $\leq$ 1.20 and gap $\geq$ 1.5. The last row is the only one where the prediction preceded the run, and it is the only miss.

Task	Fit ratio	Gap	Predicted	Outcome
dSprites	0.986×	2.68×	win	won 53.8%
dSprites + shape	1.019×	2.56×	win	won 42.9%
3D Shapes	1.457×	13.11×	loss	lost on fit
Navier-Stokes, rotation	1.337×	6.74×	loss	lost on fit
Navier-Stokes, Complex FiLM	1.122×	6.74×	win	won 64.6%
Latent editing, frozen	2.433×	2.11×	loss	lost 60.1%
PDEBench reaction-diffusion	2.381×	2815×	loss	lost to Concat-MLP
Audio effects, 4 controls	1.000×	1.13×	no gain	tied
Audio effects, 16 controls	1.009×	1.84×	win	**flat at** −1.1%

background advection is pure rotation (log-magnitude 0.000, phase 1.594). A task needing both factors is one a pure rotation cannot fit, and that is what happened, to our own arm. The rotation failed on fit while Complex FiLM, which reaches both, fit at 0.938× and won by 64.6%. The polar type named which of our arms would fail before we knew why.

The two checks are not sufficient, and the first out-of-sample test says so. We calibrated both thresholds on the comparisons they summarise, so that agreement is consistency, not predictive accuracy. We then built the case the account favors: a sixteen-control parametric chain, the high-dimensional regime where coverage arguments say structure should matter most. Both checks passed before the run, with a compositional gap of 1.83× and a fit ratio of 1.011×, so the account predicted a win. The measured advantage was flat at −1.1% across one, two, four and eight simultaneous controls.

A third condition is missing, and the instrument for it was already in the repository. A rigging check we had added only to confirm a task was not built to flatter the operator found 0 of 12 control pairs matching the operator's form. That is the class test of the previous paragraph applied to the *task* rather than to the mechanism, and it called the outcome before training. The reason is visible in hindsight: dSprites factors are independent coordinates of the generative process, so combining two changes yields a well-defined image and composition holds exactly in factor space, whereas the chain's saturation and dynamics stages interact, so the combined effect is not the composition of the individual effects. There is then nothing for exact composition to be exact about, however large the condition space or the gap. Structured conditioning pays when the task's own conditions compose the way the operator does, which is narrower than any dimensional criterion and is measurable in advance.

What to do with it. Two short runs decide the question. Fit one high-capacity conditioner, read its effective operator, and let the closed forms above give each mechanism's approximation error; measure the compositional gap on your own evaluation; then check that the task's own conditions compose the way the operator does. Reach for a group-structured conditioner when all three hold: the condition names a change to apply rather than content to produce, held-out combinations are meaningfully harder than trained ones, and the representation trains alongside the operator. Keep FiLM otherwise. Complex FiLM is the drop-in where the condition transforms content the input already supplies, at lower cost than FiLM. In rollouts, pair the operator with a latent-consistency loss, since neither works alone.

Honest status. The mathematics is elementary: Procrustes, Cauchy, and a standard Lipschitz property of the exponential on compact groups. We calibrated the thresholds quoted above on the same nine tasks they summarise, so the framework is consistent with our evidence rather than validated out of sample, and the tasks themselves were not pre-registered even though the splits were. We state the rule so that it can be falsified, and give in §8 the experiment that would do it.

8 Limitations

All evidence comes from controlled 64×64 benchmarks with known factors, width-128 models and a single conditioning site, so we validate nothing at production scale; the natural deployment experiment is the adaLN site of a text-conditioned diffusion transformer. Learned latents do not support exact group action, so the advantage on real images is the 1.8–8× error reduction quoted above rather than the $10^5\times$ of the synthetic suites, and 3D Shapes shows orthogonality can cost in-distribution fit on rich scenes. Conditions far from any group action, such as open-ended text, remain untested.

The guidance result does not transfer to a multimodal target. Our guidance benchmark measures distortion against a deterministic target, so it cannot see the distribution-sharpening CFG exists to provide. A second harness makes $p(x \mid c)$ uniform over 64 cells, so fidelity and diversity are separately measurable against a target known in closed form. An unregistered single-seed pilot on it, with no gate and no verdict claimed, goes against us on all three axes we could measure: at matched fidelity CFG retains more diversity, the strength knob is not monotone because the rotation channel wraps, and the base model fits 33% worse before any guidance at all. We therefore claim less distortion at equal cost on a deterministic target, and no more.

The experiment that would falsify the account. The rule of §7 predicts a *location*, not a direction. Build a task family whose conditional operator has a dialable stretch-to-rotation ratio, $T^*(c) = U(c)^\alpha S(c)^{1-\alpha}$; the closed forms then predict where FiLM, Complex FiLM and a rotation each take over as α sweeps. Registering those crossover points before running would convert a framework fitted on nine observations into a prediction tested out of sample, and would predict our own losses. We have not run it.

A prediction we record before testing it. World models condition on past proprioception and on past actions, usually through the same mechanism. The account says they want different ones. A proprioceptive reading names which features should be active, which is content and lives in the stretch factor, and proprioceptive states do not compose, so closure has nothing to determine; FiLM should win there. Actions compose by construction, so the group structure is real, and on our own rollout suite a composing operator with a consistency loss beat FiLM by 78% at horizon 20. The prediction is that splitting the two paths beats using one mechanism for both, that making both FiLM costs long-horizon accuracy, and that making both an operator costs single-step fit. It is a smaller intervention than anything we built, it tests the account rather than the operator, and we state it so that it can fail in public.

Reproducibility statement. We release the code, the pre-registrations, and the raw logs. Every suite has a specification committed before its decision run; a registry maps each suite label in this paper to the module that produced it, the log it wrote, and its verdict, so a replicator reproduces any row with a single command. Every number, table, and figure here regenerates from those append-only logs. Unit tests pin the operator invariants (identity at initialization and at $c = 0$, orthogonality, exact composition), the budget counters, and the registry itself. All amendments and one accounting erratum, a FLOP undercount our own audit caught that downgraded a favorable verdict, we disclose with the results.

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